Present Day, Present Time:
AI Exposure & Labor Market Risk

What's Going On

November 2022: ChatGPT launches.

Millions of people start using AI to do things that *people get paid to do*.

Some workers are much more exposed than others.

Did it cost them?

Where I got the idea: the **offshoring** literature from trade economics.

- Grossman & Rossi-Hansberg (2008): offshoring works at the *task* level
- Ottaviano, Peri & Wright (2013): different workers, different cutoffs

AI is the same story, new technology — **"offshoring" your tasks to Claudeland, GPTopia, Geminia. . .**

Toy Model Time!

A Task-Based Model (Setup)

- Tasks $i \in [0,1]$ ordered by AI susceptibility; workers of type i^* specialize in task i^*
- CES production: $Y = \left[\int_0^1 y(i)^{(\sigma-1)/\sigma}\, di\right]^{\sigma/(\sigma-1)}$
- Human supply of task i: $f(i)$ (fixed). Pre-AI wage:

$$w_0(i^*) = P_Y \left(\frac{Y_0}{f(i^*)}\right)^{1/\sigma} \qquad \text{(scarce tasks} \rightarrow \text{higher wage)}$$

- **AI enters as a supply shock:** adds $s(i) \geq 0$ to tasks below the capability frontier I^*

$$s'(i) < 0 \;\text{ for } i < I^*, \qquad s(i) = 0 \;\text{ for } i \geq I^*$$

Post-AI Wages

Effective supply $\bar{y}(i) = f(i) + s(i)$, output expands to $\bar{Y} > Y_0$.

$$w_1(i^*) = P_Y \left(\frac{\bar{Y}}{f(i^*) + s(i^*)} \right)^{1/\sigma}$$

Above I^*

$s = 0 \to$ only $\bar{Y} \uparrow$

Wages rise

Deep inside $[0, I^*)$

s/f large $\to$ denom. outpaces numer.

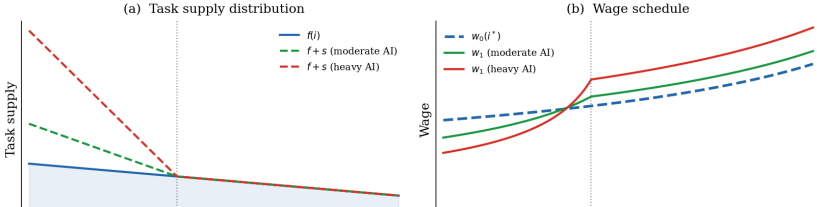
Wages fall

Near I^* from below

$s \approx 0 \to$ output growth dominates

Wages still rise

Post-AI Shock: Two Scenarios



The model says high-exposure workers should see worse outcomes post-2022.

Let's test it.

Data

CPS-ASEC (via IPUMS)

- 500K+ person-year obs, 2018–2025
- Ages 22–65, labor force participants
- Outcomes:
  - Unemployment indicator (mean 3.4%)
  - Log real annual wages (CPI-deflated)

AI Exposure (Eloundou et al. 2024)

- Occupation-level LLM exposure score (β_o)
- GPT-4 annotated, O*NET task database
- 0 (low: construction) to 0.78 (high: legal, CS)
- Time-invariant, assigned at occ. level
- Coefficients: per 0.1 increase in β_o

Empirical Strategy

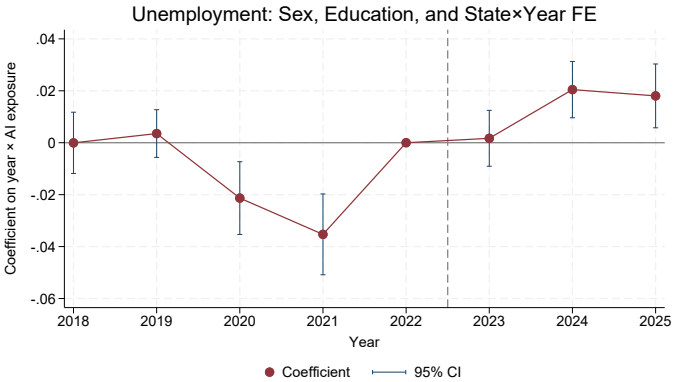
$$Y_{it} = \alpha + \sum_{t \neq 2022} \beta_t \left(\mathbf{1}[t] \times \mathrm{AIExp}_i \right) + X_{it}' \Gamma + \varepsilon_{it}$$

X_{it}: sex, education, state×year FEs
SEs clustered by occupation Baseline: 2022 ($\beta_{2022} \equiv 0$)

Four specs: (1) none → (2) sex → (3) sex+educ → **(4) + state×year**

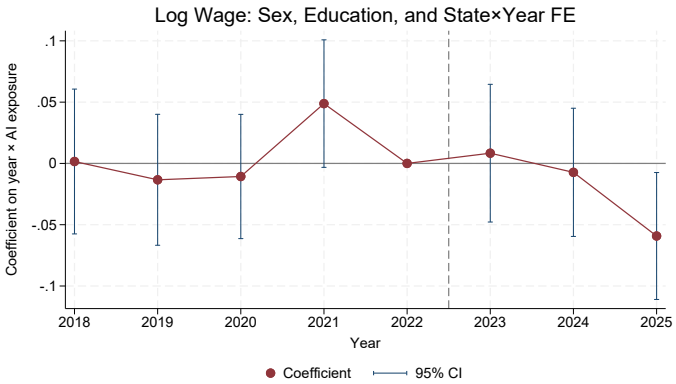
- Year-by-year interaction with AI exposure → event-study coefficients $\hat{\beta}_t$
- Pre-2022 $\hat{\beta}_t \approx 0$: parallel trends
- Post-2022 $\hat{\beta}_t$ diverges: AI effect
- **Education heterogeneity:** estimate separately for college vs. non-college

Result 1: Unemployment



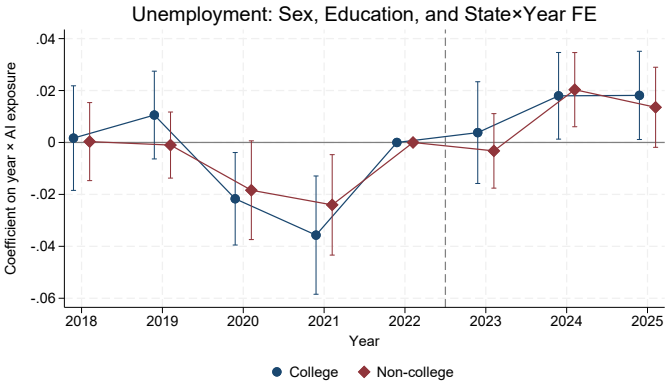
- Pre-2022: flat (2018–19)
- 2020–21: negative — AI-exposed jobs are remote-compatible, shielded from COVID layoffs
- **Post-2022: reversal**
  - 2024: $+0.021^{***}$
  - 2025: $+0.018^{***}$
- $\approx$ 6% increase relative to 3.4% baseline
- Robust across all four specs

Result 2: Wages



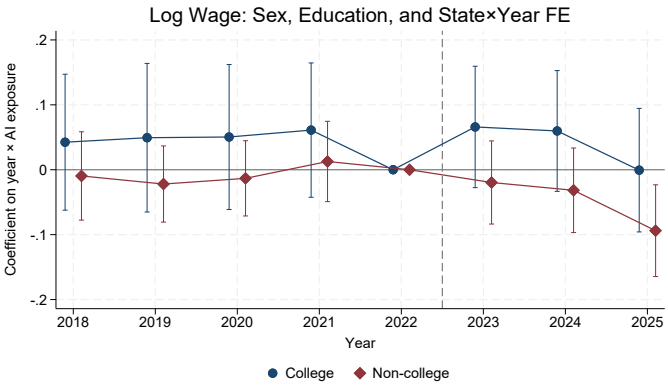
- Pre-2022: flat — clean parallel trends
- Post-2022: compression emerges late
  - 2025: -0.059^{**} (preferred spec only)
- Weaker and later than unemployment
- Not significant under looser specs
- Wage adjustment lags behind employment adjustment

Who Bears the Burden? Unemployment by Education



- **Symmetric**: both groups see comparable post-2022 unemployment increases
- College: $+0.018^{**}$ (2024, 2025)
- Non-college: $+0.020^{***}$ (2024), $+0.014^{*}$ (2025)
- CIs overlap — no significant difference
- AI unemployment risk is **broadly shared**

Who Bears the Burden? Wages by Education



- **Asymmetric**:
- College: post-2022 coefficients positive, insignificant — no compression
- **Non-college 2025:** -0.094^{***} (preferred spec)
- Grows more significant with richer FEs
- Unemployment shared; wage compression concentrated among non-college workers

Thank you!

Questions welcome.

Bottom line:

AI works like offshoring —
task-level supply shock,
heterogeneous exposure

What the data shows:

Unemployment: broad, robust, both groups
Wages: late, concentrated on non-college

Backup: The Dirty Work

Full derivation, three-group analysis, I^* discussion

Derivation: Pre-AI Equilibrium

Step 1. Cost minimization $\rightarrow$ FOC: $p_0(i) = \mu \cdot (Y/y(i))^{1/\sigma}$, $\quad \mu = P_Y$ in competitive eq.

Step 2. Task demand: $y^d(i) = Y(P_Y/p_0(i))^{\sigma}$

Step 3. Market clearing $y^d(i) = f(i)$:

$$p_0(i) = P_Y \left(\frac{Y}{f(i)} \right)^{1/\sigma}$$

Step 4. Plug $y(i) = f(i)$ into CES $\rightarrow$ output is pinned:

$$\boxed{Y_0 = \left[\int_0^1 f(i)^{(\sigma-1)/\sigma} \, di \right]^{\sigma/(\sigma-1)}}$$

Y_0 depends only on worker distribution f — not on prices.

Derivation: Post-AI Equilibrium

Step 5. AI adds supply $s(i)$. Effective supply: $\bar{y}(i) = f(i) + s(i)$.

Repeat Steps 1–4 with $\bar{y}$ in place of f:

$$\bar{Y} = \left[\int_0^1 (f(i) + s(i))^{(\sigma-1)/\sigma}\, di\right]^{\sigma/(\sigma-1)} > Y_0$$

Post-AI wage:

$$\boxed{w_1(i^*) = P_Y \left(\frac{\bar{Y}}{f(i^*) + s(i^*)}\right)^{1/\sigma}}$$

Wage ratio:

$$\frac{w_1(i^*)}{w_0(i^*)} = \underbrace{\left(\frac{\bar{Y}}{Y_0}\right)^{1/\sigma}}_{\text{output expansion}>1} \cdot \underbrace{\left(\frac{f(i^*)}{f(i^*) + s(i^*)}\right)^{1/\sigma}}_{\text{supply effect}\leq 1}$$

Three-Group Analysis (Formal)

Group 1 ($i^* \geq I^*$, $s = 0$): $w_1/w_0 = (\bar{Y}/Y_0)^{1/\sigma} > 1$ — uniform gain.

Group 2 ($i^* < I^*$, s/f large): Compression whenever $s(i^*)/f(i^*) > \bar{Y}/Y_0 - 1$.
Because $\bar{Y}/Y_0$ is a CES average over *all* tasks (most with $s = 0$), while s/f is concentrated on AI-supplied tasks, this condition holds for heavily exposed workers.

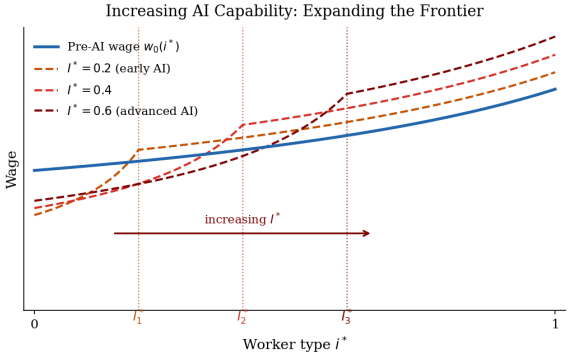
Group 3 ($i^* < I^*$, s/f small): Near I^*, $s \to 0$ so supply effect ≈ 1 and output expansion dominates $\to$ gain despite $i^* < I^*$.

Threshold $\hat{\imath} < I^*$: $s(\hat{\imath})/f(\hat{\imath}) = \bar{Y}/Y_0 - 1$.
Winners: $i^* > \hat{\imath}$. Losers: $i^* < \hat{\imath}$. The boundary lies *strictly below* I^*.

Further Discussion of I^*

As AI improves, I^* shifts right

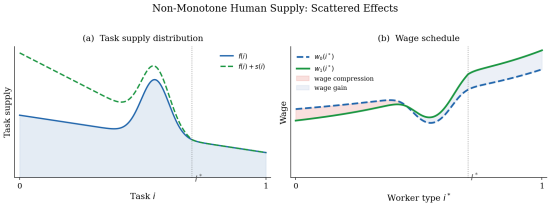
More workers exposed, deeper compression, $\hat{\imath}$ shifts rightward.



Each curve: w_1 for a different I^*. Higher I^* $\longrightarrow$ wider compression.

Non-monotone $f(i)$: scattered effects

If many workers cluster in a mid-complexity task, the local s/f ratio drops, creating pockets where workers gain even inside $[0, I^*)$.



Bump in f near $i \approx 0.55$: gain pocket inside the frontier.